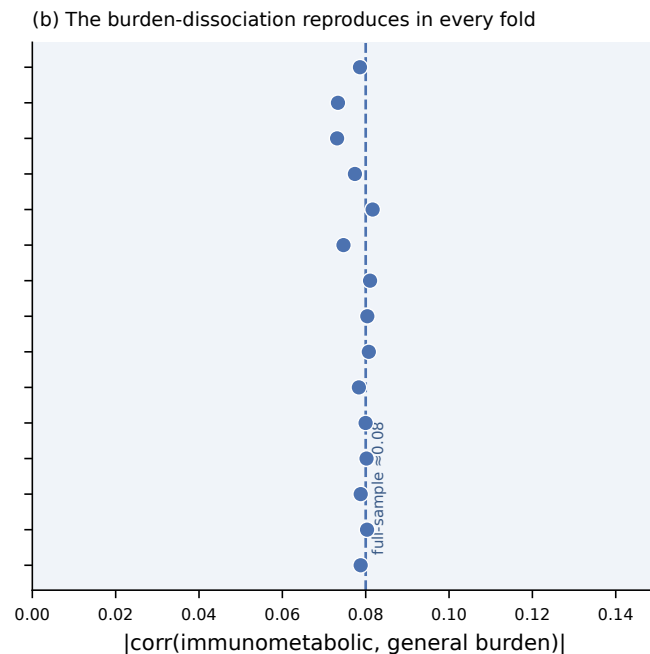
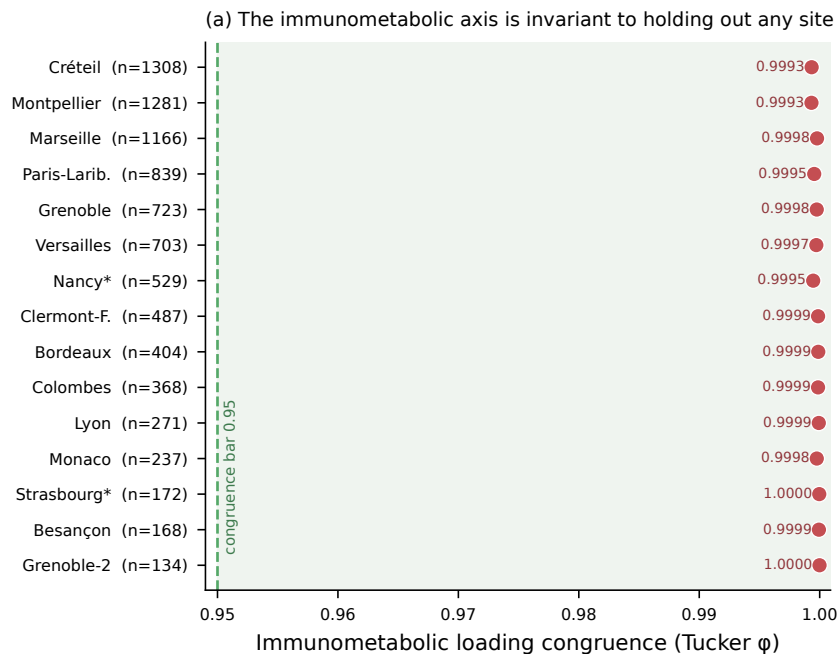


Leave-one-site-out external validity: the map and its headline finding survive out-of-site refitting



Each row is one of the 15 sites ($N \geq 100$) held out entirely; the GLLVM is refit on the remaining sites and the held-out site's loadings are projected. (a) The immunometabolic loading pattern is essentially identical whichever site is removed (Tucker ϕ 0.999–1.000, all far above the 0.95 bar). All eight factors cleared their pre-set congruence bar in all 15 folds; the single weakest factor was $\phi = 0.917$ (Monaco, $n = 237$), a thin 2–4-item axis still above the 0.85 bar for thin factors. (b) The score-level dissociation between the immunometabolic axis and general burden holds in every fold ($|r| = 0.073$ – 0.082 , vs ≈ 0.08 on the full sample) — the finding is not driven by any single site.